CCC-C-446F
July 23 , 1986
SUPERSEDING
CCC-C-446E
August 15, 1969

FEDERAL SPECIFICATION

CLOTH, MUSLIN, COTTON

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton muslin cloth (see 6.1).

1.2 Classification.

1.2.1 Types and classes. The cloth shall be of the types as specified in table I and in the following classes as specified (see 6.2):

Class 1 - Unbleached
Class 2 - Bleached or dyed
Class 3 - Bleached or dyed; preshrunk

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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Federal Specifications

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Federal Standards

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Laws and Regulations

Federal Trade Commission

Rules and Regulations Under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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Color Association of the United States

Standard Color Reference of America

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016.)

American Association of Textile Chemists and Colorists (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard sample. Unless otherwise specified (see 3.3.3), the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced (see 6.3).

3.2 Material. Recycled material may be used (see 6.5).

3.2.1 Yarn. The yarn shall be made from cotton that has been carded or combed, drawn and spun into single yarns for both warp and filling.

3.3 Color. The color shall be natural, bleached white or dyed as specified (see 6.2). When the following colors are specified by civil agencies and standard samples are not available, the colors shall approximate the cable numbers of the Standard Color Reference of America as listed:

Color	Cable Number
Scarlet	80080
Jasmine	80067
Princeton Orange	80070
Old Glory Blue	80075
Old Glory Red	80108
Bleach White	80001

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3.3.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

3.3.2 Matching. The color and appearance of the dyed and finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and have a correlated color temperature of 7000 ± 500K, with illumination of 100 ± 20 foot candles. The color shall be a good match to the standard sample under incandescent lamplight at 2300 ± 100K.

3.3.3 Colorfastness. The dyed and finished cloth shall show colorfastness to laundering (after 3 cycles), light, and perspiration equal to or better than the standard sample or equal to or better than a rating of "good" when tested as specified in 4.2.3. The cloth shall show colorfastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating not lower than 3.5 for all colors except Blacks, Navy Blues, and Reds, which shall have an AATCC Chromatic Transference Scale rating not lower than 1.5, when tested as specified in 4.2.3.

3.4 Spectral reflectance for Brown 436, class 3. When specified (see 6.2), the finished cloth shall meet the spectral reflectance factors (in percent) for the visible near infrared wavelength range 600 to 860 nanometers (nm) as specified in table I, when tested as specified in 4.2.3.

TABLE I. Spectral reflectance requirements for Brown 436 (class 3)

Wavelength (nanometers)	Reflectance (percent) Max	Min	Wavelength (nanometers)	Reflectance (percent) Max	Min
600	20	12	740	62	20
620	20	12	760	65	20
640	20	12	780	67	20
660	26	12	800	69	20
680	35	15	820	71	20
700	46	20	840	73	20
720	56	20	860	75	20

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3.5 Physical requirements. The finished cloth shall conform to the requirements in table II when tested as specified in 4.2.3.

TABLE II. Physical requirements

Type	Class	Weight per square yard (ounces), minimum	Yarns per inch, minimum		Breaking strength (pounds), minimum	
			Warp	Filling	Warp	Filling
I	1	2.7	64	58	37	26
	2	2.4	66	52	34	20
	3	2.5	68	56	36	22
II	1	3.1	68	70	42	34
	2	2.7	70	64	38	28
	3	2.8	72	68	40	30
III	1	3.4	78	76	46	39
	2	2.9	80	70	45	32
	3	3.0	83	74	44	34
IV	1	2.4	38	36	37	20
V	1	3.9	56	60	44	50
VI	2	2.4	64	56	35	25
VII	2	3.1	85	72	45	30

3.5.1 Weave. The weave shall be plain.

3.5.2 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless looms, the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

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3.6 Finish (see 6.1). Class 2 and 3 cloth shall be singed.

3.6.1 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material for class 1 cloth shall not exceed 12.0 percent when tested as specified in 4.2.3. The starch and protein content including chloroform-soluble and water-soluble material for classes 2 and 3 cloth shall not exceed 2.0 percent when tested as specified in 4.2.3.

3.6.2 pH. The pH value of the water extract of the dyed and finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.2.3.

3.7 Dimensional stability. The shrinkage or elongation in both the warp and filling of the finished cloth shall not be greater than 9.0 percent for class 1 cloth, 6.0 percent for class 2 cloth, and 2.0 percent for class 3 cloth, when tested as specified in 4.2.3. The preshrinkage process used shall not be identified by name or trademark, on either the cloth, ticket, or the package.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in 5.1.

3.9 Fiber identification. Each piece shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.10 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "FACE" at each end of the piece.

3.11 Workmanship. The cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each piece in the sample shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage of the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section I, FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the maximum limits shown on Sears Fabric Detect Scales (see 6.6) F or 2, as applicable, for slubs and D or 3 1/2, as applicable, for knots shall be scored. No linear yard from any piece within the sample shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards exceed the following values:

30.0 points for classes 2 or 3 dyed cloth
35.0 points for classes 2 or 3 white cloth
45.0 points for class 1 cloth

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The lot shall be unacceptable if two or more individual pieces exceed the following values:

45.0 points for classes 2 or 3 dyed cloth
53.0 points for classes 2 or 3 white cloth
68.0 points for class 1 cloth

If no individual piece exceeds the point level, the lot shall be accepted. If one piece exceeds the point level, a second sample of 20 pieces shall be selected for individual piece quality examination only. If no piece in the second sample exceeds the point level, the lot shall be accepted. If one or more pieces exceed the level, the lot shall be rejected. Point computation for lot quality and individual piece quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width less than specified
Poor dye penetration, mottled, streaky or cloudy (when applicable)
Excessive neppiness

4.2.2.2 Length examination.

4.2.2.2.1 Individual pieces. During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

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4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.2.3 Shade and appearance examination. During the yard-by-yard examination, each piece in the sample shall be examined for shade and appearance. Any piece in the sample off shade, shaded side to side, side to center, or end to end, or any piece in the sample that does not have an appearance equal to or better than the standard sample shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Roll identification examination. During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample have the "FACE" stamping missing from either or both ends or the "FACE" stamping is on the wrong side; contain identification of the preshrinkage process by name or trademark on the cloth or ticket; or are not labeled, ticketed or invoiced in accordance with the Textile Fiber Products Identification Act.

4.2.3 End item testing. The methods of testing specified in FED-STD-191 whenever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of yards. The sample unit shall be 3 1/2 continuous yards full width, for classes 1, 2 and 3 cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.2.1	1200 1/
Presence of labile sulfur	3.3.1	2020 1/
Colorfastness to:		
Laundering after 3 cycles	3.3.3	5610 2/
Light	3.3.3	5660
Perspiration	3.3.3	5680
Crocking	3.3.3	5651
Spectral reflectance	3.4	4.3.1
Weight, ozs sq yd	3.5	5041
Yarns per inch	3.5	5050
Breaking strength	3.5	5100
Weave	3.5.1	Visual 3/
Finish:		
Singed (classes 2 and 3)	3.6	1/
Nonfibrous material	3.6.1	2611
pH	3.6.2	2811
Dimensional stability	3.7	5550

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ The specimens must be dried after each of the three laundering cycles. Only the stains on the cotton fibers shall be evaluated.

3/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

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4.3 Method of inspection.

4.3.1 Spectral reflectance measurements in the visible/near infrared. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm), at 20 nm intervals on a spectrophotometer (see 6.7) relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white; e.g., Halon, magnesium oxide, or vitrolite tiles (see 6.8). The spectral bandwidth shall be less than 25 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CIE Source A or CIE Source D65. The specimen shall be measured as a single layer, backed with three layers of the same fabric and shade. Measurements will be taken on a minimum of two different areas, and the data averaged. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent, and wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. When the measured reflectance values for any color at four or more wavelengths do not meet the limits specified in table I, it shall be a test failure.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Levels A, B, or Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

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6. NOTES.

6.1 Intended use. The cloth covered by this specification is intended for use in clothing, flags, and equipage items for personnel of the Department of Defense and civil Government personnel. Users with special requirements such as flame resistance should submit information on intended use to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2.1).
c. Color required (see 3.3).
d. When spectral reflectance is required for class 3 cloth (see 3.4).
e. Width of cloth required (see 3.5.2).
f. When length is other than specified (see 3.8).
g. Selection of applicable levels of put-up, preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to samples, address the procuring office issuing the invitation for bids.

6.4 Dye combinations. One of the following suggested, but not mandatory, dye combinations for Shade Brown 436 may be used:

Combination 1	Combination 2	Combination 3
Reactive Yellow 37	Reactive Yellow 84	Reactive Orange 91
Reactive Red 180	Reactive Scarlet H-EA	Reactive 183
Reactive Blue 19	Reactive Blue 160	Reactive Blue 184
Rodacyl Orange 26		

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.2).

6.6 Fabric defect scales. Fabric Defect Replica Kits are available from:

Sears Roebuck and Company
Department 817, Attn: BSC 23-29
Sears Tower
Chicago, IL 60684

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6.7 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR, and Macbeth 1500 with IR option.

6.8 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Bureau of Standards or the instrument manufacturers.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review activities

Army - MD
Navy - MC
DLA-CT

User activities

Army - IH
Air Force - 82

PREPARING ACTIVITY:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
JUS-FPI

Project No. 8305-0082

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
CCC-C-446F	Cloth, Muslin, Cotton

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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